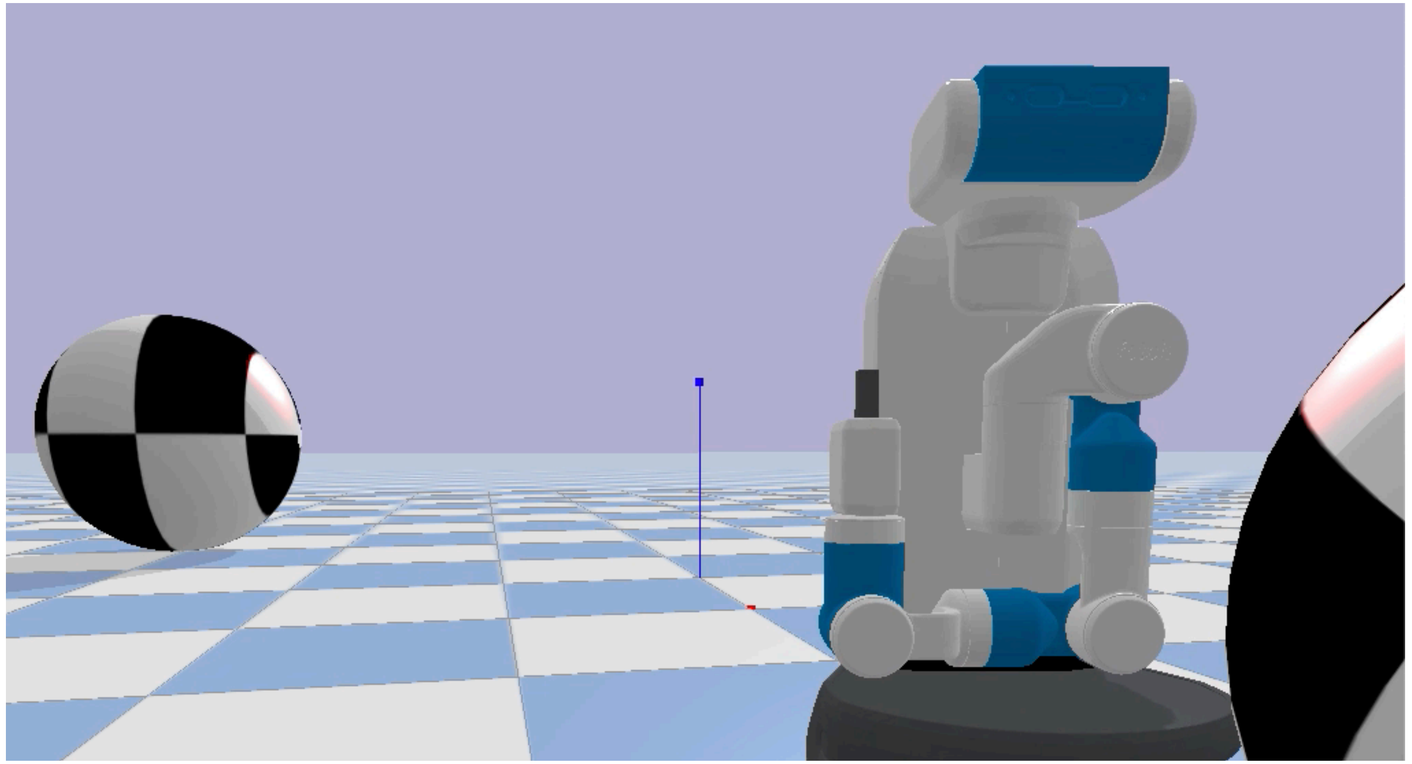
ROB-UY 3303

Robot Motion & Planning

Lecture 06A - Path & Traj Tracking



Agenda

1. Path Tracking Control
2. Trajectory Tracking Control via Points
3. Trajectory Tracking Control via Acceleration

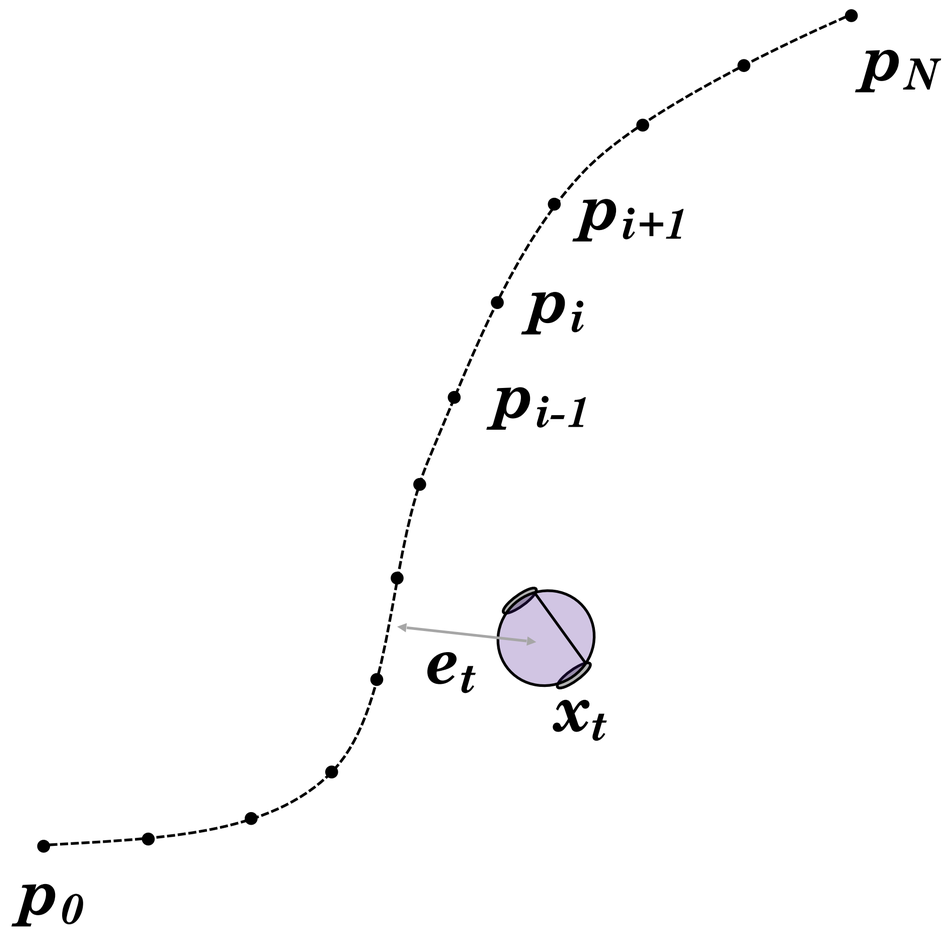
Agenda

1. **Path Tracking Control**
2. Trajectory Tracking Control via Points
3. Trajectory Tracking Control via Acceleration

- Assume kinematic model with control signals v, w :

$$\begin{bmatrix} \dot{x}(t) \\ \dot{y}(t) \\ \dot{\theta}(t) \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 \\ \sin\theta & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v(t) \\ w(t) \end{bmatrix}$$

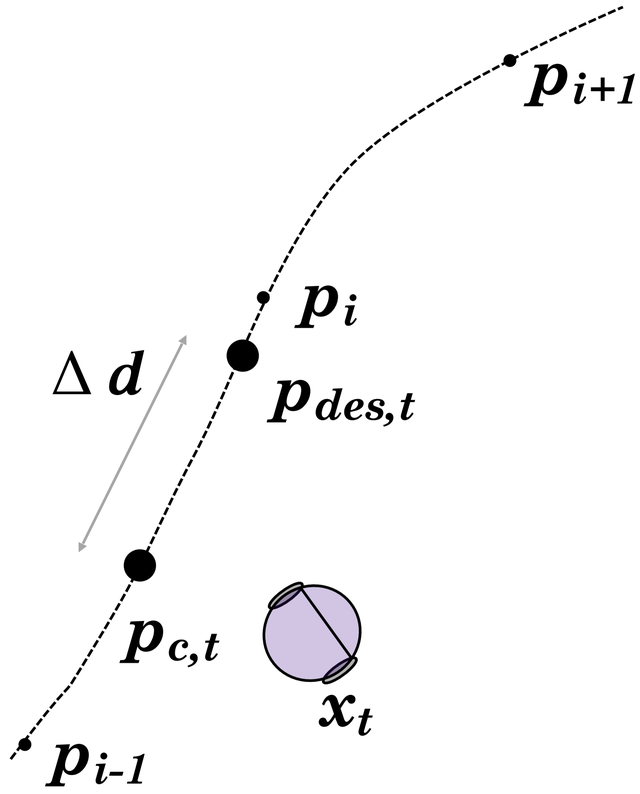
- **Problem**
  - Given
    - A sequence of robot poses (e.g. $\boldsymbol{p_i} = [\, x\ y\ \theta]_i$) defining a path $\boldsymbol{P}$
    - The robot's current pose at time step t (e.g. $\boldsymbol{x_t} = [\, x\ y\ \theta]_t$)
  - Output
    - Robot control signals at time step t (e.g. $\boldsymbol{u_t} = [v\ w]_t$) that minimize tracking error (e.g. $\Sigma t e_t = \Sigma t\ f(\boldsymbol{x}_t, \boldsymbol{P})$ where f returns the shortest distance from $\boldsymbol{x}_t$ to path $\boldsymbol{P}$)



- Assume we have a point tracking controller with:

$$v = k_\rho \rho \qquad w = k_\alpha \alpha + k_\beta \beta$$

- **Solution**
  - At each time step t:



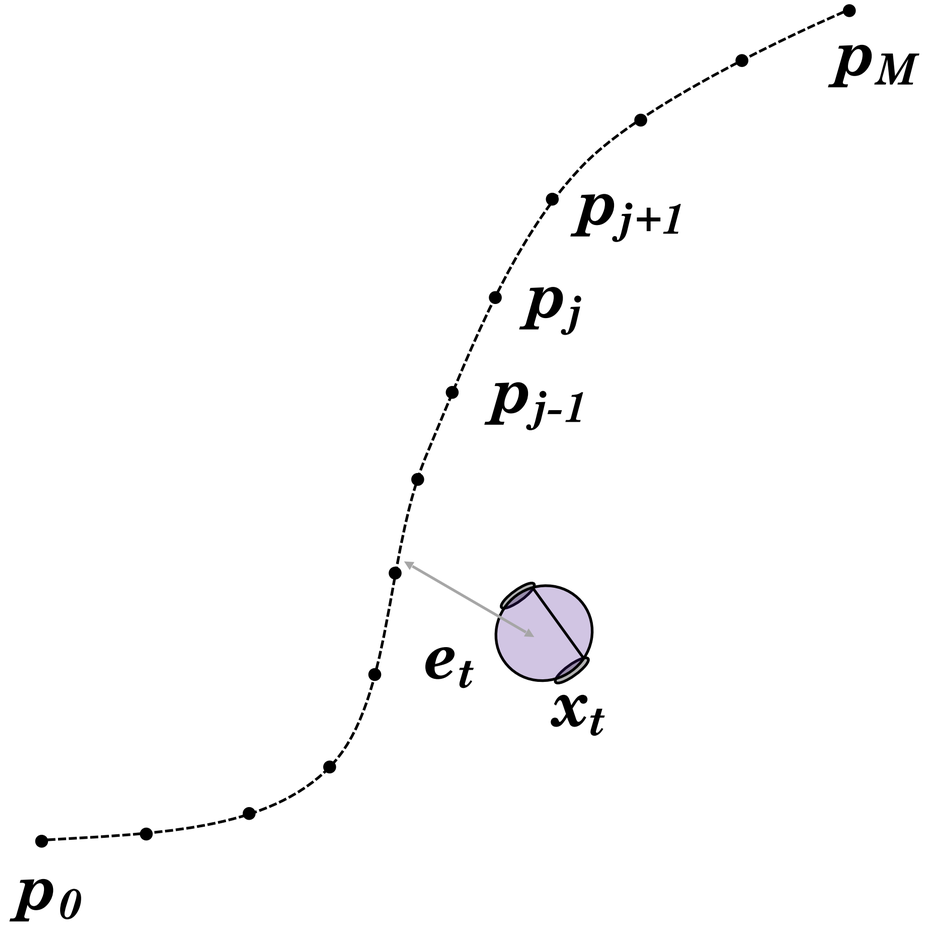
Agenda

1. Path Tracking Control
2. **Trajectory Tracking Control via Points**
3. Trajectory Tracking Control via Acceleration

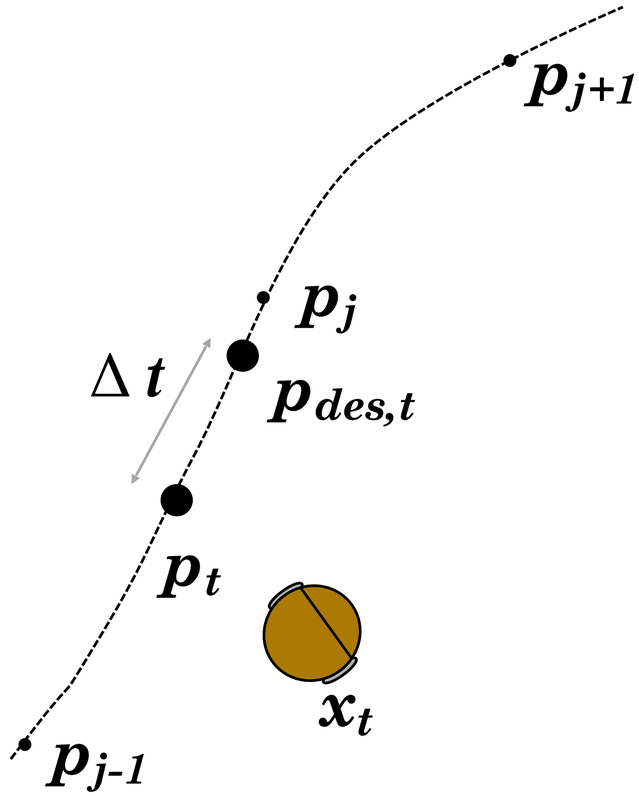
- Assume kinematic model with control signals v, w :

$$\begin{bmatrix} \dot{x}(t) \\ \dot{y}(t) \\ \dot{\theta}(t) \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 \\ \sin\theta & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v(t) \\ w(t) \end{bmatrix}$$

- **Problem**
  - Given
    - A sequence of trajectory points (e.g. $\boldsymbol{p_j} = [t\ x\ y\ \theta]_j$) defining a trajectory $\boldsymbol{T}$
    - The robot's current pose at time step t (e.g. $\boldsymbol{x_t} = [t\ x\ y\ \theta]_t$)
  - Output
    - Robot control signals at time t (e.g. $\boldsymbol{u_t} = [v\ w]_t$) that minimize tracking error (e.g. $\Sigma t\ e_t = \Sigma t\ g(\boldsymbol{x_t}, \boldsymbol{T})$ where g returns the distance from $\boldsymbol{x_t}$ to the point on $\boldsymbol{T}$ with the same timestamp)



- **Solution**
  - At each time step t:
    1. Find the pose $\boldsymbol{p}_t$ on traj $\boldsymbol{T}$ that has the same timestamp as $\boldsymbol{x}_t$.
    2. Find the pose $\boldsymbol{p}_{des,t}$ on traj $\boldsymbol{T}$ that is a time $\Delta \boldsymbol{t}$ further along the path from pose $\boldsymbol{p}_t$.
    3. Calculate control signals v, w using pose $\boldsymbol{p}_{des,t}$ as an input to the point tracking controller.



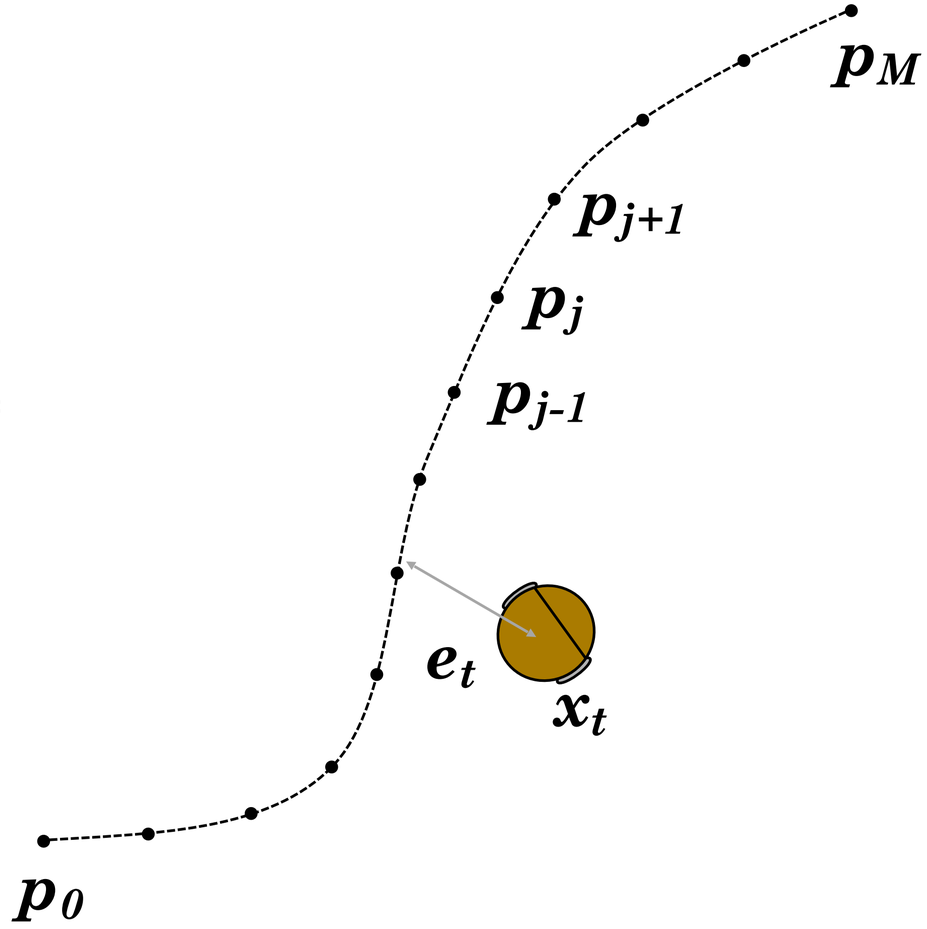
Agenda

1. Path Tracking Control
2. Trajectory Tracking Control via Points
3. **Trajectory Tracking Control via Acceleration**

- Assumed model with control inputs a, w :

$$\begin{bmatrix} \dot{x}(t) \\ \dot{y}(t) \\ \dot{v}(t) \\ \dot{\theta}(t) \end{bmatrix} = \begin{bmatrix} v\cos\theta \\ v\sin\theta \\ a \\ w \end{bmatrix}$$

- **Problem**
  - Given
    - A sequence of trajectory points (e.g. $\boldsymbol{p_j}$ = $[t\ x\ y\ \theta\ \dot{x}\ \dot{y}\ \dot{\theta}\ \ddot{x}\ \ddot{y}\ \ddot{\theta}]_j$,) defining a trajectory $\boldsymbol{T}$
    - The robot's current pose at time step t (e.g. $\boldsymbol{x_t}$ = $[t\ x\ y\ \theta\ \dot{x}\ \dot{y}\ \dot{\theta}\ \ddot{x}\ \ddot{y}\ \ddot{\theta}]_t$)
  - Output
    - Robot control signals at time t (e.g. $\boldsymbol{u_t} = [a\ w]_t$) that minimize tracking error (e.g. $\Sigma t\ e_t = \Sigma t\ g(\boldsymbol{x}_t, \boldsymbol{T})$ where g returns the distance from $\boldsymbol{x}_t$ to the point on $\boldsymbol{T}$ with the same timestamp)



- **Solution**
  - And the control law…

$$\begin{bmatrix} a \\ \omega \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -V\sin(\theta) \\ \sin(\theta) & V\cos(\theta) \end{bmatrix}^{-1} \begin{bmatrix} \ddot{x}_d - k_{px}(x - x_d) - k_{dx}(\dot{x} - \dot{x}_d) \\ \ddot{y}_d - k_{py}(y - y_d) - k_{dy}(\dot{y} - \dot{y}_d) \end{bmatrix}$$



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